

Trainings ▾

General Information

WARNING

Compressed natural gas is normally treated with an odor producing chemical so that users will be able to smell gas leaks. Always be alert for the smell of gas. If you enter a room or approach a vehicle and a smell of gas is present, immediately shut off all engines and ignition sources. Avoid sparks, arcing switches and equipment, cigarettes, pilot lights, flames, and other sources of ignition in the area and in areas with common ventilation. Provide extra ventilation to the area and do not start the equipment or nearby equipment until the leak is corrected and the area is ventilated. Avoid leaving natural gas fueled equipment in unventilated rooms overnight or for extended periods. Store and service natural gas fueled equipment in large well-ventilated areas or outside.

WARNING

If natural gas leaks are present, do not store the vehicle inside or in any area that is covered. Severe personal injury can result from asphyxiation or explosion.

WARNING

Natural gas is highly flammable. Keep all cigarettes, sparks, arcing switches and equipment, pilot lights, flames and other sources of ignition out of the work area and areas sharing ventilation.

WARNING

Not all types of Natural Gas are treated with an odorant. Gas leaks from a non-refined source, such as Landfill Gas, Biogas, Coal Bed Gas, or Wellhead Gas, can not always be detected by smell.

WARNING

Do not troubleshoot or repair gas leaks while the engine is running.

CAUTION

Natural gas is lighter than air and can accumulate under the hood and awnings.

⚠ CAUTION ⚠

Always tighten fasteners and fuel connections to the required specifications. Overtightening or under tightening can allow leakage. These connections are critical to the fuel and air systems.

⚠ CAUTION ⚠

Always test for fuel leaks as instructed, as odorant can fade.

⚠ CAUTION ⚠

Close the manual fuel valves prior to performing maintenance and repairs, and when storing the vehicle inside.

Specifications

This section presents the specifications for natural gas engines.

Cummins® spark-ignited engines that use natural gas as a fuel source provide a low emission alternative for various applications. In order for the engines to continually provide extremely low emission levels and provide the best durability and reliability, Cummins Inc. has developed several fuel standards. Cummins® Engineering Standard (CES) 20067, Natural Gas Fuel; CES 14604, 14624, Natural Gas Fuel; and CES 14608 Wide Range Natural Gas Fuel define some of the natural gas specifications.

These specifications apply to fuel as it is delivered to the engine, regardless of whether its origin was liquid or gaseous. Liquefied natural gas (LNG) is an acceptable fuel, provided the onboard fuel storage and supply system delivers proper pressure, temperature, and complete vaporization to the engine fuel system inlet. These specifications are **not** intended to cover certification requirements. **Only** engines that have been specifically designed, built, and approved by Cummins Inc. for use with alternative fuels (including Landfill and Biogas) can be operated with low energy or aggressive fuels. The fuel **must not** contain water, dust, sand, dirt, oils, or any other substance or component in an amount that is detrimental to the operation of the engine. The use of gas with chlorine additives is **not** permitted. More specifications and test methods are detailed in these standards. For alternative fuels information, contact a Cummins® Authorized Repair Location for assistance.

Engine type (rich-burn, lean-burn, or alternative fuel) and application (automotive, industrial, or generator) may dictate unique fuel requirements. The applicable engine operation and maintenance (O&M) manual **must** be referenced for the required fuel specifications. Operators of Cummins® natural gas engines should communicate the specifications to potential fuel suppliers and request confirmation as to local availability.

For CES 20067, the basic chemical composition for natural gas is detailed in Table 13, CES 20067 Chemical Composition. The Wobbe index **must** be between 1300 and 1377 as measured by ASTM D3588. The Wobbe index is a calculated value. Reference CES 20067 for more information.

Table 13, CES 20067 Chemical Composition

Constituents	Requirements	Test Method
Methane (CH ₄)	90.0 percent volume minimum	ASTM D1945
Ethane (C ₂ H ₆)	4.0 percent volume maximum	ASTM D1945
Propane (C ₃ H ₈)	1.7 percent volume maximum	ASTM D1945
Butane and Heavier (C ₄ H ₁₀ ⁺)	0.7 percent volume maximum	ASTM D1945
Carbon Dioxide and Nitrogen (CO ₂ + N ₂)	3.0 percent volume maximum	ASTM D1945
Hydrogen (H ₂)	0.1 percent volume maximum	ASTM D2650
Carbon Monoxide (CO)	0.1 percent volume maximum	ASTM D2650
Oxygen (O ₂)	0.5 percent volume maximum	ASTM D1945
Sulfur (S)	0.001 percent weight maximum	Title 17 CCR Section 94112 Method 16

Table 14 below shows the applicable standard for common on-highway Cummins® natural gas engines.

Table 14, Fuel Standards for Cummins® Natural Gas Engines

Standard	Engine Family			
	B5.9 G, C8.3 G, B6.7 G	ISB5.9 G, B Gas International, B Gas Plus, C Gas Plus, L Gas Plus	ISL G, ISX12 G	B6.7N, L9N, ISX12N, X15N
CES 14604 Minimum Methane Number: 80 Minimum Higher Heating Value: 975 BTU/Standard Cubic Feet	Yes	N/A	N/A	N/A

Table 14, Fuel Standards for Cummins® Natural Gas Engines

Standard	Engine Family			
	B5.9 G, C8.3 G, B6.7 G	ISB5.9 G, B Gas International, B Gas Plus, C Gas Plus, L Gas Plus	ISL G, ISX12 G	B6.7N, L9N, ISX12N, X15N
CES 14624 Minimum Methane Number: 75 Minimum Lower Heating Value: 44194 kJ/kg [19,000 BTU/lbm]	N/A	N/A	Yes	Yes
CES 14608 Minimum Methane Number: 65 Minimum Lower Heating Value: 37448.6 kJ/kg [16100 BTU/lbm]	N/A	Yes	N/A	N/A

For CES 14604, CES 14624, and CES 14608, the basic chemical composition for natural gas is detailed in Table 14, CES 14604, CES 14624, and CES 14608 Chemical Composition. Further information on each of the standards follows Table 15.

Table 15, CES 14604, CES 14624, and CES 14608 Chemical Composition

Constituents	Test Method
Methane (CH ₄)	ASTM D1945
Ethane (C ₂ H ₆)	ASTM D1945
Propane (C ₃ H ₈)	ASTM D1945
Butane and Heavier (C ₄ H ₁₀ +))	ASTM D1945
Carbon Dioxide and Nitrogen (CO ₂ + N ₂)	ASTM D1945
Hydrogen (H ₂)	ASTM D2650
Carbon Monoxide (CO)	ASTM D2650
Oxygen (O ₂)	ASTM D1945
Sulfur (S)	Title 17 CCR Section 94112 Method 16
Halogens	ASTM D7449

For CES 14604, the methane number shall **not** be below 80 and the higher heating value shall **not** be below 975 BTU/Standard Cubic Feet. The methane number and higher heating value are calculated values. The maximum allowable sulfur content shall **not** be greater than 0.001 percent weight. Reference CES 14604 for more information. CES 14604 applies to B5.9 G and C8.3 G engines.

For CES 14624, the methane number shall **not** be below 75 and the lower heating value shall **not** be below 19,000 BTU/lbm. The methane number and lower heating value are calculated values. Reference CES 14624 for more information. Table 16, CES 14608 and CES 14624 Maximum Allowable Halogens, Hydrogen, Hydrogen Sulfide, Sulfur, and Siloxanes, specifies that there are four constituents in the natural gas mixture that **must** meet certain requirements to be used in the ISB6.7 G, B6.7N, ISL G, L9N, ISX12N, and ISX12 G engines.

For CES 14608, the methane number shall **not** be below 65 and the lower heating value shall **not** be below 16100 BTU/lbm. The methane number and lower heating value are calculated values. Reference CES 14608 for more information. Table 16, CES 14608 and CES 14624 Maximum Allowable Halogens, Hydrogen, Hydrogen Sulfide, Sulfur, and Siloxanes, specifies that there are four constituents in the natural gas mixture that **must** meet certain requirements to be used in ISB5.9 G, B Gas International, B Gas Plus, C Gas Plus, and L Gas Plus engines.

Table 16, CES 14608 and CES 14624 Maximum Allowable Halogens, Hydrogen, Hydrogen Sulfide, Sulfur, and Siloxanes

Constituents	Requirements	Test Method
Hydrogen (H ₂)	0.03 percent volume maximum	ASTM D2650
Hydrogen Sulfide (H ₂ S)	0.0006 percent volume maximum	ASTM D4084
Siloxanes	0.0003 percent volume maximum	EPA TO-14 15 GC/ELCD, GC/AED, GC/MS, ASTM D8230
Sulfur (S)	0.001 percent weight maximum	Title 17 CCR Section 94112 Method 16
Halogens	1.0 Parts per Million by weight (ppm(w)).	ASTM D7994

Contact a Cummins® Authorized Repair Location for information regarding calculating methane number, higher heating value, and lower heating values. The following is an example using CES 14604 to determine if the fuel is compliant.

Test Fuel Data Input (See Notes at Right)

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Location (Description)

Methane	CH ₄	90.20	%
Ethane	C ₂ H ₆	4.03	%
Propane	C ₃ H ₈	1.76	%
Butane	C ₄ H ₁₀	0.01	%
Pentane	C ₅ H ₁₂	0.01	%
Hexane	C ₆ H ₁₄	0.00	%
Heptane	C ₇ H ₁₆	0.00	%
Octane	C ₈ H ₁₈	0.00	%
Carbon Dioxide	CO ₂	0.00	%
Nitrogen	N ₂	3.99	%
Oxygen	O ₂	0.00	%
Sum of Components		100.00	%

NOTES:

1. Fuel requirements for automotive spark-ignited gas engines only.
2. Fuel as delivered to engine regardless if liquid or gaseous.
3. Maximum allowable sulfur content = 0.001% Weight.
4. Fuel shall not contain water, dust, sand, dirt, oils or any substance that can harm engine operation.

Methane Number:

89.76

PASS

(Minimum Methane Number: 80)

Higher Heating Value (BTU/Std. Cu. Ft.):

1024.50

PASS

(Min. Higher Heating Value = 975 BTU/Std. Cu. Ft.)

NOTE: Both MN and HHV Criteria Must be Met to Pass a Given Fuel!

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Cummins® natural gas engines are designed and adjusted to meet performance and emissions standards with fuel meeting these specifications. The engine may operate on fuels possessing a wide range of properties, but performance and emissions will be affected. In extreme cases, fuel with characteristics outside of these specifications can cause engine reliability or durability issues. Cummins Inc. Engine Warranty covers malfunctions that are a result of defects in material or factory workmanship. Engine damage, service issues, and/or performance issues determined by Cummins Inc. to be caused by the use of fuel **not** meeting these specifications are **not** considered to be defects in material or workmanship and are **not** covered under Cummins Inc. engine warranty.

Operators **must** be alert for sudden changes in engine operation, power levels, or the presence of knock. Each of these symptoms can be a sign of substandard fuel. If an issue related to fuel quality is suspected, ask the fuel supplier to sample and analyze the fuel in the vehicle or the fuel being supplied to the engine in stationary applications. Contact a Cummins® Authorized Repair Location for information regarding calculating methane numbers, heating values, and for further assistance.

Fuel Filters

⚠ CAUTION ⚠

Gas is extremely flammable. Contents under pressure. Vent gas from the filter by opening the drain on the filter.

⚠ CAUTION ⚠

Overtightening will distort the filter cartridge, damage the filter seal, or crack the filter head. Do not use a filter element that has been dented or damaged prior to, or during, installation.

⚠ CAUTION ⚠

Oil getting inside of the gas mass flow sensor or on the screen pack will cause poor performance.

Fuel filters are required equipment on all Cummins® natural gas engines, except the ISX12N. They are designed to remove oil and harmful particles from the fuel before they damage the fuel system or other engine components. These filters are a coalescent type filter that will capture the oil contaminants and moisture typically found in natural gas.

Oil can be introduced into a natural gas engine's fuel system in several ways. The most common is from the fueling station compressor. The oil can also be present in the station tank from the refining process. Oil in the fuel will cause the gas mass flow sensor and the heated oxygen sensor to read incorrectly. Engine performance will be affected.

The fuel filters, Cummins Filtration™ NG 5900 and NG5910, need to be drained as part of the daily or refueling maintenance check. The interval period for draining the fuel filter is dependent on the fueling station and varies for each location. The drain interval **must** be adjusted to the time required to accumulate no more than 30 milliliters [1 ounce] of oil in the fuel filter or daily, whichever occurs first.

Reference the engine operation and maintenance manual for fuel filter replacement intervals.